TEXORA · LECTURE SHEET

Carbohydrates

Sugars and their polymers: classification, the structure of glucose in Fischer and Haworth forms, anomers and mutarotation, the classical structure proofs, reducing and non-reducing sugars, sucrose inversion, and starch versus cellulose — the chemistry of cotton itself.

CHE-1205 · CHEMISTRY-II · TOPIC 6 · LEVEL 1 TERM 2

1. Definition and classification

Carbohydrates are polyhydroxy aldehydes or ketones (or substances hydrolysing to them). The old formula $C_n(H_2O)_n$ fits the common sugars but is only a nickname — the structural definition above is the real one.

CLASS	DEFINITION	EXAMPLES
Monosaccharides	Cannot be hydrolysed further; classified as aldoses/ketoses and by chain length	Glucose, fructose, galactose (hexoses, $C_6H_{12}O_6$)
Disaccharides	Two monosaccharide units joined by a glycosidic bond	Sucrose, maltose, lactose ($C_{12}H_{22}O_{11}$)
Polysaccharides	Long chains of monosaccharide units	Starch, cellulose, glycogen — all $(C_6H_{10}O_5)_n$, yet utterly different

The **D/L** label reads the –OH on the highest-numbered chiral carbon in a Fischer projection: right = D (nearly all natural sugars), left = L. D-glucose is the reference sugar of biology.

2. Glucose — open chain, ring, and the two anomers

In solution glucose is less than 0.1% open-chain aldehyde; it cyclises (C-5 hydroxyl onto C-1) into a six-membered **pyranose** ring. Cyclisation makes C-1 a new stereocentre — the **anomeric carbon** — giving two **anomers**: α (anomeric –OH trans to the CH_2OH group; down in a standard Haworth projection of a D-sugar) and β (cis; up). Freshly dissolved α-D-glucose rotates light +112°, β-D-glucose +19°; either solution drifts to the same equilibrium value, +52.7°, as anomers interconvert through the open form — **mutarotation**, with the equilibrium mixture ≈ 36% α and 64% β. The classical open-

chain structure rests on five proofs: HI/red P reduces glucose to n-hexane (a straight six-carbon chain); bromine water oxidises it gently to gluconic acid (an aldehyde group, both products six-carbon); hot HNO_3 gives saccharic acid, a *di*carboxylic acid (so the other end is CH_2OH); acetic anhydride gives a penta-acetate (five –OH groups); HCN adds on (a carbonyl is present).

3. Fructose and the osazone link

Fructose is a ketohexose (C-2 ketone), strongly laevorotatory (−92°), cyclising mainly into a five-membered **furanose** ring. With phenylhydrazine, glucose, fructose *and* mannose all give the **same osazone** (yellow crystals, mp 205 °C) — the reaction remodels only C-1 and C-2, and the three sugars are identical from C-3 down. Fructose, though a ketose, reduces Tollens' and Fehling's reagents: in the alkaline test solution it rearranges (Lobry de Bruyn) to an aldose mixture — a standard exam trap with a standard explanation.

4. Reducing sugars and the disaccharides

A sugar is **reducing** if it has a free anomeric carbon (a hemiacetal that can open to an aldehyde). Hence glucose, fructose, maltose and lactose all reduce Fehling's solution. **Sucrose** does not: its glycosidic bond joins glucose's anomeric C-1 to fructose's anomeric C-2 — *both* anomeric carbons are locked, no ring can open, no reduction. Hydrolysis (acid or invertase) frees them, and the rotation flips from +66.5° to −19.7° — the average of glucose (+52.7°) and the much more laevorotatory fructose (−92°) — hence the name **invert sugar** for the hydrolysate, sweeter than sucrose itself and the basis of syrups and soft-centre chocolates. **Maltose** links two glucoses $\alpha(1\rightarrow4)$, leaving one anomeric carbon free (reducing); **lactose** links galactose $\beta(1\rightarrow4)$ to glucose (reducing; the milk sugar).

5. Starch and cellulose — one monomer, two worlds

Both are glucose polymers; the linkage stereochemistry decides everything. **Starch** uses $\alpha(1\rightarrow4)$ bonds (amylose, a coiling chain that traps iodine blue-black) plus $\alpha(1\rightarrow6)$ branch points (amylopectin); human enzymes cut α links, so starch is food. **Cellulose** uses $\beta(1\rightarrow4)$ bonds: the chain runs straight, neighbouring chains hydrogen-bond into cables of enormous strength, and no human enzyme hydrolyses the β link — cellulose is fibre, not food. Cotton is ≈ 90% cellulose; every topic in textile engineering that touches scouring, mercerising or dye uptake is, at bottom, the chemistry of this polysaccharide's β-linked glucose chains and their hydroxyl groups.

6. Key points

- Carbohydrates = polyhydroxy aldehydes/ketones; mono-, di- and polysaccharides by hydrolysis depth.

- Glucose cyclises to a pyranose; α/β anomers at C-1 mutarotate to +52.7° (36% α, 64% β).
- Same osazone from glucose, fructose, mannose — they differ only at C-1/C-2.
- Reducing power needs a free anomeric carbon: sucrose (both locked) is the famous non-reducer; its hydrolysis is inversion.
- α links make starch digestible food; β links make cellulose structural fibre — and cotton.

7. Practice

Answer in your own words; brief model answers follow.

1. A freshly prepared solution of α-D-glucose shows rotation +112°, falling over time to +52.7°. Name the phenomenon and explain it, given β-D-glucose is +19°.
2. Why do glucose, fructose and mannose give the identical osazone with phenylhydrazine?
3. Sucrose does not reduce Fehling's solution, but after boiling with dilute acid it does. Explain both observations.
4. Verify the inversion figure: hydrolysis turns sucrose (+66.5°) into an equimolar glucose–fructose mixture. Given glucose +52.7° and fructose −92°, find the mixture's rotation.
5. Starch and cellulose have the same empirical formula and the same monomer, yet one is food and the other is the fibre of cotton. Explain in terms of the glycosidic bond.
6. Glucose is oxidised (a) with bromine water and (b) with concentrated HNO_3. Name both products and state what each proves about the structure.

Answer key

1. Mutarotation. In solution the ring opens to the free aldehyde and recloses either way, so α and β anomers interconvert until equilibrium (36% α, 64% β). The rotation of the mixture, $0.36 \times 112 + 0.64 \times 19 \approx +52.7°$, is reached from either pure anomer.
2. Osazone formation involves only C-1 and C-2 (the carbonyl carbon and its neighbour are both converted to phenylhydrazone units, destroying their stereochemical differences). Glucose, fructose and mannose have identical configurations at C-3, C-4 and C-5, so one osazone serves all three.
3. Sucrose's glycosidic bond joins the anomeric carbons of both units; neither ring can open to a free aldehyde, so Fehling's solution is untouched. Acid hydrolysis breaks the bond, releasing glucose and fructose — both reducing sugars — so the hydrolysate reduces Fehling's solution (and Tollens').
4. Equimolar mixture: $(+52.7 + (-92))/2 = -19.65 \approx -19.7°$. The sign of rotation inverts from +66.5° to −19.7°, which is why hydrolysis of sucrose is called inversion and the product invert sugar.
5. Starch's glucose units are joined α(1→4) (with α(1→6) branches); the α link gives a bent, coiling chain that human amylases hydrolyse — an energy store. Cellulose's β(1→4) links give straight

chains that pack and hydrogen-bond into strong fibres, and humans have no cellulase to cut them. Same monomer, opposite stereochemistry at one bond — two different material worlds.

6. (a) Gluconic acid — a monocarboxylic acid of six carbons: the oxidised group was an aldehyde at a chain end. (b) Saccharic (glucaric) acid — a dicarboxylic acid: the far end of the chain was a primary alcohol, $–CH_2OH$, oxidisable under the stronger conditions. Together they fix glucose as a straight-chain aldohexose, $HOCH_2–(CHOH)_4–CHO$.
